A method for producing solitary water waves and measuring their interaction with a ramp.

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We experimentally investigate soliton wave production in a wave flume and their breaking on a plane ramp of slope $s = 1/27.1$. The KdV speed relation $c' = 1 + \eta'$ is confirmed at water depths of 85 mm and 90 mm, with $P \approx 0.4$, while the sech^2 wave profile at $h = 85$ mm yields $A = 0.997 \pm 0.012$ and $B = 1.078 \pm 0.017$ in good agreement with theory. Experimental breaking distances agree qualitatively with Grilli et al. (1997) but are systematically a factor of ≈ 1.5–2 smaller than predicted, attributed to non-adiabatic energy loss at the ramp surface and a bias in the experimental breaking criterion.

I. INTRODUCTION

Solitary waves were first documented by the Scottish engineer John Scott Russell in 1834 while observing a boat that had come to an abrupt stop in a narrow channel [1]. The waveform he observed maintained its shape for "one or two miles" and appeared to be above the undisturbed surface with no clear trough. In 1895, Korteweg and de Vries deduced a non-linear equation which models a generalised case of solitary waves called solitons [2]. The non-dimensional KdV equation can be written as follows,

$$\frac{\partial \eta}{\partial t} + \eta \frac{\partial \eta}{\partial x} + \frac{\partial^3 \eta}{\partial x^3} = 0, \tag{1}$$

where η is the height of the wave above the undisturbed equilibrium [3]. If we then assume a standard travelling soliton solution of the form $\eta\,(x,t) = a\ sech^2(x - ct)$, we can substitute into the dimensional KdV equation to find

$$c = \sqrt{gh}\left(1 + \frac{\eta_0}{2h} + \alpha\left(\frac{\eta_0}{h}\right)^2 + \dots\right), \tag{2}$$

where c is the speed of the soliton moving with peak height η_0 in water of depth h and α is a positive constant. If we then assume we are working in the regime where $\eta_0/h \ll 1$, known as weak nonlinearity, we can neglect terms $\mathcal{O}\left(\eta_0/h\right)^2$ and we arrive at the relation

$$c = \sqrt{gh}\left(1 + \frac{\eta_0}{2h}\right). \tag{3}$$

It is useful to define new parameters $c' = c/\sqrt{gh}$ and $\eta' = \eta_0/2h$ which lead to the following relation.

$$c' = 1 + \eta'. \tag{4}$$

Equation (4) can be used to experimentally determine if a water wave exhibits a soliton behaviour according to its peak height and speed in a given body of water.

Soliton solutions to the KdV equation have a particular nondimensionalised shape (at $t = 0$) given by

$$H'(X) = \text{sech}^2(X), \tag{5}$$

where $H' = \eta/\eta_0$, $X = x\sqrt{3\eta_0/4h^3}$.

We can also consider the propagation of solitary waves through a changing water depth in the form of a ramp. Instead of propagating with invariant shape, a soliton travelling up a ramp exhibits shoaling - where the wave height increases and the wavelength decreases. If the ramp is steep enough relative to the dimensions of the soliton, the effect of shoaling causes the wave to break. We will adopt the definition of wave breaking proposed by Grilli et al. in 1997; the break point of a wave is the location where the wave front face has a vertical tangent [4].

FIG. 1 shows the relevant parameters in the study of solitons breaking on a ramp. We follow the analysis of Grilli et al. [4] by defining the dimensionless quantity $H_0' = \eta_0/h_0$ which leads to our first criterion. Note that H_0' is distinct from the profile normalisation $H' = \eta/\eta_0$ and here h_0 is the undisturbed water depth (equivalent to h in earlier equations).

For a wave to break on a ramp at θ from the horizontal, we must have

$$H_0' > 16.9\ s^2, \tag{6}$$

where $s = \tan\theta$.

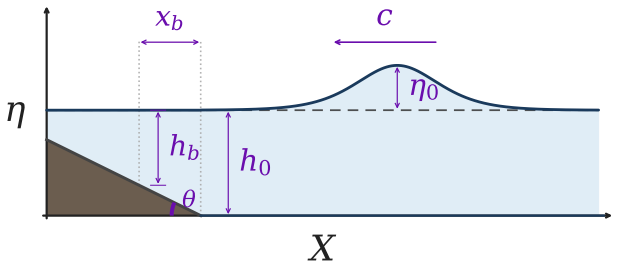


FIG. 1: Diagram showing key parameters.

We define two more parameters:

$$L_0 = \frac{2h_0}{\sqrt{3H_0'/4}}\text{arctanh}\left(\sqrt{3}/3\right), \tag{7}$$

the characteristic wave length scale from Boussinesq theory, and

$$S_0 = \frac{sL_0}{h_0} = 1.521\frac{s}{\sqrt{H_0'}}, \tag{8}$$

a scaled slope parameter.

Grilli et al. found that between $0.025 < S_0 < 0.30$ waves exhibit "plunging" - where the crest overtakes the wave, pitches over and falls into the water. If dealing with plunging waves, we can find the breaking depth

$$\frac{h_b}{h_0} = \frac{0.149}{(S_0/H_0')^{0.523}}, \tag{9}$$

which leads to the breaking distance

$$x_b = \frac{h_0 - h_b}{s}. \tag{10}$$

Since solitary waves have been found to successfully model tsunami waves, experiments to investigate their wave breaking on shorelines have far reaching applications [5].

II. METHODS

We begin with an analysis of the validity of our experimental method of producing solitons. **FIG. 2** shows the experimental setup. Our first hypothesis was that soliton waves could be produced by pushing the paddle by hand from a horizontal distance of 5cm to the latch. Note that in our initial experiment we used an empty tank with the ramp removed.

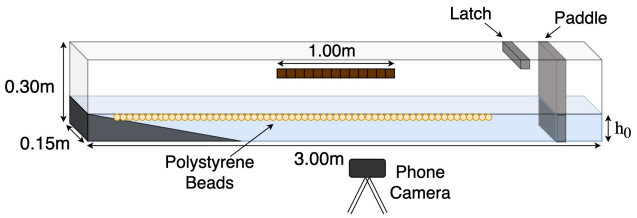


FIG. 2: Experimental setup for the soliton interacting with a ramp. The experimental setup for our initial experiment was identical with the ramp removed.

Before each wave, orange polystyrene beads were arranged along the surface of the undisturbed water, against the nearside tank wall, to make the wave profile clearer on camera. As described, the paddle was pushed by hand to create wave excitations, and the propagating waves were filmed using a phone camera on a stand. The phone camera was positioned approximately level with the water level in each run to reduce parallax error. The tank was filled to water depths ranging from 50 ± 1mm to 90 ± 1mm in steps of 5mm, each measured using a ruler held against the inside of the tank wall. At each water depth, four or five waves were created and filmed.

A spirit level was used to ensure the tank was horizontal. A ruler was strapped to the tank horizontally to provide calibration for the motion tracking software. The pixel-to-metre calibration used was 0.579 mm/px. White sheets of paper were taped to the back of the tank to increase the contrast between the beads and the background.

Each video was analysed using the motion-tracking software "Tracker". In each frame, the wave peak was marked at the bead with the highest vertical displacement. The software then extracted the speed c and the vertical displacement η_0 of the wave as it propagated down the tank. **FIG. 3** shows an example of the frame by frame tracking.

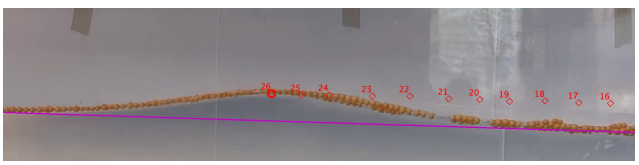


FIG. 3: A frame from the Tracker software.

Our second hypothesis was that our experimental setup, including the ramp, could demonstrate wave breaking according to the various predictions from theory [4]. The ramp used was 0.15 ± 0.01m wide (since tape was used to cover the gap to the tank wall), 39 ± 1mm tall and 1.05 ± 0.01m long. In this experiment, the water depths were $55, 60, 65, 70, 75, 80$ and 90 ± 1mm with five waves filmed at each depth. All other experimental methods were identical between the two investigations.

The wave was determined to break when at least one pair of beads was arranged in a vertical line. This is an approximation of the previously discussed definition of wave breaking from the theory [4]. The distance x_b was obtained by manually marking the horizontal position at which the wave broke and using the inbuilt measurement function in Tracker.

III. RESULTS

TABLE I: Minimised χ^2 fit parameters for $c' = a + b\eta'$ at each water depth h_0. Theory predicts $a = 1$ and $b = 1$.

h_0 (mm)	$a \pm \sigma_a$	$b \pm \sigma_b$	$\tilde{\chi}^2$	$P(\chi^2, \nu)$
50	0.739 ± 0.107	1.566 ± 0.352	2.08	0.101
55	1.025 ± 0.056	0.737 ± 0.181	5.29	0.001
60	0.953 ± 0.135	0.959 ± 0.545	0.81	0.487
65	1.104 ± 0.057	0.319 ± 0.264	0.70	0.553
70	1.076 ± 0.053	0.326 ± 0.305	2.53	0.055
75	0.904 ± 0.102	1.366 ± 0.602	1.78	0.149
80	1.055 ± 0.077	0.495 ± 0.557	7.80	0.000
85	0.949 ± 0.032	1.060 ± 0.235	1.13	0.336
90	0.997 ± 0.038	0.842 ± 0.254	0.99	0.398

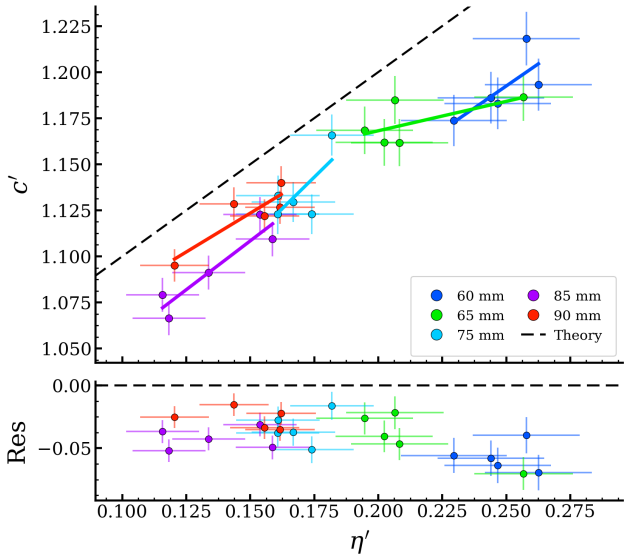


FIG. 4: Scaled wave speed c' against scaled wave peak height η' for five water depths. The theory line $c' = 1 + \eta'$, error bars and normalised residuals are shown.

TABLE I shows the minimised χ^2 fit parameters for all nine depths with values for $\tilde{\chi}^2$ and $P(\chi^2, \nu)$.

FIG. 4 shows data for the five depths with reduced χ^2 values in the range $0.70 \leq \tilde{\chi}^2 \leq 2.00$, selected as those showing the strongest agreement with the KdV speed relation. Solid lines show the minimised χ^2 linear fits for each data point. Normalised residuals are shown below the graph. We note that all data points lie under the theory line $c' = 1 + \eta'$. This systematic offset is discussed in Section IV. As predicted by theory - due to the nondimensional scaling of c and η - we can also see a tendency for data points to aggregate closer to the origin as the water depth increases.

FIG. 5 shows a plot of the wave profile from one video at a water depth of $h = 85$ mm. The predicted wave profile

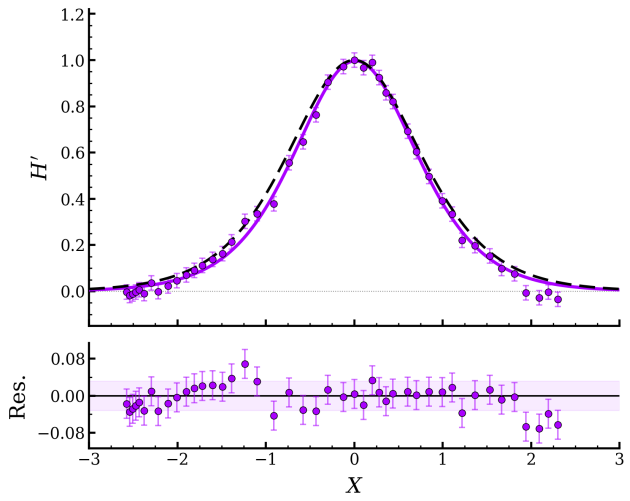


FIG. 5: Scaled wave profile H' against scaled horizontal position X for a single wave at $h = 85$mm. Error bars and residuals are shown.

of a soliton from equation 5 is shown. The data were fit to $H' = A\ \mathrm{sech}^2(B \cdot X)$ to obtain $A = 0.997 \pm 0.012$, $B = 1.078 \pm 0.017$ with a reduced chi squared of $\tilde{\chi}^2 = 0.93$. $P(\chi^2, \nu) = 0.601$ for this fit. The error bars show $\sigma_{H'} = 0.0417$ and since $\sigma_X = 6.75 \times 10^{-3}$ the horizontal error bars are not visible. The purple band on the residual plot shows the region $\pm\sigma_{H'}$.

TABLE II: Predicted breaking distances x_b from equation (10) compared to experimental measurements for each water depth h_0. The ramp length is 1055 mm.

h_0 (mm)	η_0 (mm)	x_b Grilli (mm)	x_b Expt. (mm)
90	39.0	1586	did not break
80	40.0	1321	817 ± 10
75	42.0	1164	618 ± 10
70	42.5	1034	568 ± 10
65	40.0	952	468 ± 10
60	38.0	861	395 ± 10
55	38.0	740	341 ± 10

TABLE II shows the results of our second experiment. We see that our experimental values of x_b are roughly half the length of theoretical predictions. In contrast to theory, waves at water depths of 75mm and 80mm were observed to break on the ramp of length 1055mm. The experimental breaking distance x_b carries a larger uncertainty than the pixel calibration alone would suggest, since the breaking criterion — at least one pair of beads arranged vertically — is a subjective visual judgement applied to a moving video frame. We estimate this introduces an uncertainty of approximately $\pm$10mm.

IV. DISCUSSION

We begin by analysing the agreement of our data to the soliton model from equation (4). The depths 55mm and 80mm yielded $P < 0.01$ indicating that the scatter in the data is inconsistent with the stated uncertainties, suggesting unaccounted systematic errors at these depths. This could be explained by a particularly large unaccounted for variation in the force with which the paddle was manually pushed between runs.

The depths 65mm and 70mm give poor agreement in $b = 1$ with values 2.6σ and 2.2σ away. At these depths, the range of η' sampled was small, meaning the data does not provide meaningful constraints on the value of b.

Data from a depth of 50mm gave a fit with relatively poor agreement in both a and b that resulted in a noticeably steeper linear relationship. Since the KdV model, equation (3), is most accurate in the regime where $\eta_0 \ll h_0$, at $h_0 = 50$mm and $\eta_0 \approx 38$mm this approximation becomes invalid. Such a steep relationship indicates that the nonlinear terms $\mathcal{O}\left(\eta_0/h\right)^2$ that were neglected in the KdV derivation of equation (3) now give a meaningful contribution to the wave speed at this depth.

Water depths of 60mm and 75mm show good agreement with theory - values for a and b are within 1σ - however, we note that the uncertainties at this depth were relatively large. Data from depths of 85mm and 90mm are in excellent agreement with theory and show strongly linear relationships with $P \approx 0.4$. In the interest of legibility, we chose to select the five depths for which $0.70 \leq \tilde{\chi}^2 \leq 2.00$ to include in **FIG.4**.

As previously mentioned, the data from our first experiment is systematically below the predicted theory, with all residuals being negative. The consistency of this negative bias across all five selected depths suggests a systematic physical effect rather than random experimental error. Physically, this could be the result of viscous damping effects. In real fluids, waves lose energy due to viscous dissipation at the tank walls and bottom. This results in a reduced wave speed relative to the inviscid KdV model [6].

Possible improvements to our initial experiment include: using a motorised paddle to make reproducible waves, using a larger tank to increase the tracking region and gather more data on c' and η', using capacitance wave gauges, and using smaller beads to better approximate the surface of the water. In particular, we could use a motorised paddle to create a nonlinear displacement of water via the Goring algorithm, which has been shown to reliably produce soliton waves [7]. Using capacitance gauges to measure η_0 has been shown to greatly increase the accuracy of measurement [8].

FIG. 5 shows good agreement between our wave profile and the form of a soliton. The fit parameter $A = 0.997 \pm 0.012$ is in excellent agreement with theory ($A = 1$), suggesting that the waves produced have peak heights consistent with soliton waves. The fit parameter $B = 1.078 \pm 0.017$ is 4.6σ from the predicted value, indicating that the waves produced were narrower than is expected for soliton waves. This could be due to the wave not yet fully forming into a soliton profile. Alternatively, viscous dissipation would also produce an artificially narrower measured profile. Both $\tilde{\chi}^2 = 0.930$ and $P = 0.601$ suggest the model accurately describes our data, and hence our first hypothesis - that our methodology can produce soliton waves - can be accepted for water depths of 85mm and 90mm.

Sample wave profiles at shallower depths showed much poorer agreement with equation (5) mainly due to depressions in the tail of the wave below the undisturbed water level. A key property of soliton waves is that the wave profile lies entirely above the undisturbed water level, and so these troughs were a clear indication of non-soliton behaviour. As discussed earlier, at these shallow depths the waves produced are outside of the regime assumed for the

KdV model to be valid. All improvements to the previous experiment would apply to the wave profile experiment.

Turning to our ramp experiment, we first note that our data shows a clearly monotonically increasing x_b with h_0 which is in qualitative agreement with theory. However, we see a consistent discrepancy between the magnitudes where theory predicts $\approx 1.5-2$ times longer breaking distances than we observed. Since this trend is present across all water depths, it is likely that there are systematic sources of energy loss in our experimental setup. While the gaps between the ramp and the tank were covered by tape, there were small leaks which acted as potential areas for water to escape and dissipate energy. We also note that the surface of the ramp likely had a higher coefficient of friction than the perspex surface of the tank which would clearly increase energy dissipation.

The Grilli et al. (1997) model assumes adiabatic shoaling; that the wave amplitude grows quasi-statically as depth decreases, conserving wave energy at each point. This requires the depth to vary slowly relative to the wavelength of the wave. However, as the wave travelled from the smooth perspex tank floor onto the rougher surface of the ramp, the sudden increase in bed friction could have caused a rapid, non-adiabatic loss of energy, taking the wave outside the regime assumed by the Grilli model and causing it to break earlier than predicted.

The experimental breaking criterion — at least one pair of beads aligned vertically — is likely a systematic source of error that causes x_b to be underestimated relative to the theoretical definition. The beads were not perfectly evenly distributed before each wave, were not of uniform shape or size, and there was slight surface turbulence present. Since only a single pair of beads needs to satisfy the vertical alignment condition, random perturbations caused by these factors could trigger the criterion before the wave front has developed a true vertical tangent. This would cause the experimental breaking point to be identified systematically earlier than the theoretical one, producing a consistent negative bias in the measured x_b values across all depths, consistent with the factor of $\approx 1.5-2$ discrepancy observed in **TABLE II**.

All improvements to the previous experiments would apply to our final experiment. Further improvements include using a smoother ramp surface that is flush with the tank walls, using a gentler slope to better approximate adiabatic shoaling, and using smaller depth increments. Other experimental criteria for determining wave breaking could be implemented to better approximate the definition - that breaking occurs where the wave front face has a vertical tangent. Fitting the bead positions to a continuous line and searching for vertical tangents could yield better agreement with theoretical predictions, for example. A higher frame rate would also be useful to increase the number of frames from which wave breaking can be determined.

V. CONCLUSIONS

We have investigated experimentally the production of soliton waves in a wave flume and their interaction with a ramp, testing two hypotheses against the predictions of KdV theory and the Grilli et al. (1997) breaking criterion.

For our first hypothesis, we fitted the dimensionless speed relation $c' = a + b\eta'$ to data at nine water depths. At depths of $85\,\text{mm}$ and $90\,\text{mm}$, we find a and b consistent with the theoretical values of unity, with $P \approx 0.4$, confirming that our paddle methodology can produce soliton waves at these depths. At shallower depths the KdV approximation breaks down as the nonlinearity parameter η_0/h approaches unity, and the data departs systematically from the theoretical predictions. A systematic negative offset in c' relative to the KdV prediction was observed across all depths, attributed to viscous damping not accounted for in the inviscid KdV model. The dimensionless wave profile at $h = 85\,\text{mm}$ was well described by $H' = A\,\text{sech}^2(BX)$ with $A = 0.997 \pm 0.012$ and $B = 1.078 \pm 0.017$, giving $\tilde{\chi}^2 = 0.93$ and $P = 0.60$. This further suggests soliton behaviour at these depths.

For our second hypothesis, the experimental breaking distances x_b increase monotonically with water depth h_0, in qualitative agreement with the Grilli et al. (1997) model. However, the measured values are systematically a factor of ≈ 1.5–2 smaller than the theoretical predictions across all depths. We attribute this discrepancy primarily to a sudden, non-adiabatic energy loss as the wave transitions from the smooth perspex tank floor to the rougher ramp surface, and to a systematic bias in the experimental breaking criterion which identifies breaking earlier than the theoretical definition of a vertical wave front tangent [4].

Future work should focus on using a motorised wavemaker with Goring's algorithm [7] to produce more reproducible solitons, capacitance wave gauges to measure η_0 more precisely [8], and a smoother ramp to better satisfy the adiabatic shoaling assumption of the Grilli model.

REFERENCES

[1] John Scott Russell. Report on Waves: Made to the Meetings of the British Association. 1845.

[2] Dr. D. J. Korteweg and Dr. G. de Vries. On the change of form of long waves advancing in a rectangular canal, and on a new type of long stationary waves. 1895.

[3] J. E. Allen. The Early History of Solitons (Solitary Waves). Physica Scripta, 57(3):436–440, 1998. doi: 10.1088/0031-8949/57/3/016.

[4] S. T. Grilli, I. A. Svendsen, and R. Subramanya. Breaking Criterion and Characteristics for Solitary Waves on Slopes. Journal of Waterway, Port, Coastal, and Ocean Engineering, 123(3):102–112, 1997. doi: 10.1061/(ASCE)0733-950X(1997)123:3(102).

[5] C. E. Synolakis. The Run-up of Solitary Waves. Journal of Fluid Mechanics, 185:523–545, 1987. doi: 10.1017/S002211208700329X.

[6] L. Fernand Mouassom, T. Nkoa Nkomom, Alain Mvogo, and Cesar Biouele Mbane. Effects of viscosity and surface tension on soliton dynamics in the generalized KdV equation for shallow water waves. Communications in Nonlinear Science and Numerical Simulation, 102:105942, 2021. doi: 10.1016/j.cnsns.2021.105942.

[7] Derek G. Goring. Tsunamis: The Propagation of Long Waves onto a Shelf. Technical Report KH-R-38, W.M. Keck Laboratory of Hydraulics and Water Resources, California Institute of Technology, Pasadena, CA, 1978.

[8] Hung-Chu Hsu, Yang-Yih Chen, and Hwung-Hweng Hwung. Experimental study of the particle paths in solitary water waves. Philosophical Transactions of the Royal Society A, 370(1964):1629–1637, 2012. doi: 10.1098/rsta.2011.0445.

[9] Ifan Hughes and Thomas Hase. Measurements and their Uncertainties. Oxford University Press, 2010.

ERROR APPENDIX

The analysis below follows the convention of Hughes and Hayes [9].

Experiment 1: Wave Speed

The water depth h_0 at each run was measured using a ruler held against the inside of the tank wall, with an uncertainty of $\sigma_{h_0} = \pm 1$ mm.

The wave speed c was extracted in Tracker by fitting a straight line to the horizontal peak position versus time over $N \approx 35$ frames at a frame rate of $f = 30$ fps, giving a total tracking time of $T = N/f \approx 1.17$ s. A metre ruler was used to calibrate distances on the videos with 1px $\approx 5.794 \times 10^{-4}$m. The uncertainty in locating the wave crest at each frame is estimated as $\sigma_{\text{px}} = 2$ px, corresponding to $\sigma_l = 1.16$ mm. Propagating this through the gradient gives

$$\sigma_c = \frac{\sqrt{2}\,\sigma_l}{T} = \frac{\sqrt{2} \times 1.16 \times 10^{-3}}{1.17} \approx 0.0014\ \text{m s}^{-1}. \tag{11}$$

We use the conservative value $\sigma_c = 0.002\,\text{m s}^{-1}$ throughout the analysis.

The wave amplitude η_0 was determined from the maximum vertical displacement of the peak bead above the undisturbed baseline. Since both the peak position and the baseline are each uncertain by σ_l, the uncertainty in η_0 is $\sigma_{\eta_0} = \sqrt{2}\,\sigma_l = \approx 2.4$ mm.

The dimensionless parameters c' and η' are defined as

$$c' = \frac{c}{\sqrt{g h_0}}, \qquad \eta' = \frac{\eta_0}{2h_0}. \tag{12}$$

Their uncertainties were propagated using the standard formula. For c':

$$\frac{\sigma_{c'}}{c'} = \sqrt{\left(\frac{\sigma_c}{c}\right)^2 + \left(\frac{1}{2}\frac{\sigma_{h_0}}{h_0}\right)^2}, \tag{13}$$

and similar for η'.

The data at each depth were fitted to $c' = a + b\eta'$ by minimising χ^2 with weights $w_i = 1/\sigma_{c'_i}^2$ using the weighted normal equations. The fit parameters and their uncertainties were extracted from the covariance matrix as discussed in Chapter 7 of Hughes and Hayes [9]. The resulting values of $a \pm \sigma_a$, $b \pm \sigma_b$, $\tilde{\chi}^2$ and $P(\chi^2, \nu)$ are shown in **TABLE I**.

Experiment 2: Wave Profile

The pixel-to-metre calibration was 0.579 mm/px. The dominant source of uncertainty was the precision with which bead centres could be located in ImageJ, estimated as $\sigma_{\text{px}} = 1.5$ px. This gives a uniform uncertainty on the normalised surface elevation:

$$\sigma_{H'} = \frac{\sigma_{\text{px}} \times 0.579 \times 10^{-3}}{\eta_0} = 0.0313, \tag{14}$$

and on the dimensionless horizontal coordinate $X = x\sqrt{3\eta_0/4h^3}$:

$$\sigma_X = \sigma_{\text{px}} \times 0.579 \times 10^{-3} \times \sqrt{\frac{3\eta_0}{4h^3}} = 5.55 \times 10^{-3}. \tag{15}$$

Since $\sigma_X \ll \sigma_{H'}$, the horizontal error bars are not visible in **FIG. 5**. A camera tilt correction was applied by fitting a straight line through points in the undisturbed flat water regions at either end of the bead array, and subtracting this baseline from all vertical positions.

The data were fitted to $H' = A\,\text{sech}^2(B \cdot X)$ using χ^2 minimisation in Python via scipy.optimize.curve.fit. The total uncertainty on each point was computed as

$$\sigma_{\text{total}} = \sqrt{\sigma_{H'}^2 + \left(\left|\frac{d\,\text{sech}^2}{dX}\right| \sigma_X\right)^2}, \tag{16}$$

where the derivative term accounts for the propagation of horizontal uncertainty into the fitted value. This is called the derivative rule in Hughes and Hayes [9]. The fit yielded $A = 0.997 \pm 0.012$, $B = 1.078 \pm 0.017$, with $\tilde{\chi}^2 = 0.93$ and $P(\chi^2, \nu) = 0.601$.

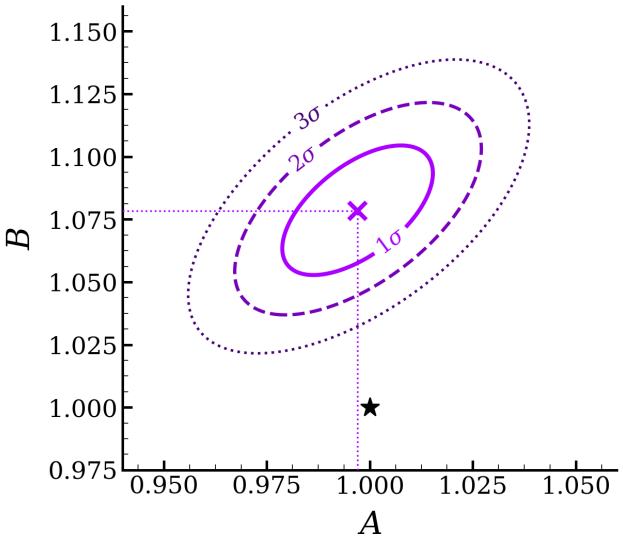


FIG. 6: χ^2 contour graph showing the minimisation to find $A = 0.997$ and $B = 1.078$. The star shows the position of the theoretical prediction while the cross shows the minimised position.

The χ^2 contour plot shows the theory point ($A = 1$, $B = 1$) lying well outside the 3σ contour around the minimised values, with $\Delta\chi^2 = 38.2$ relative to the minimum. This at first appears alarming however, this must be interpreted in the context of the large number of degrees of freedom $\nu = 41$ — giving $\tilde{\chi}^2 = 0.93$, which confirms a good fit to the data.

Experiment 3: Wave Breaking

The ramp dimensions were measured with a ruler: height 39 ± 1 mm, length 1055 ± 10 mm, giving a slope $s = 39/1055 = 0.0370 \pm 0.001$. The water depths were measured using a ruler held against the inside of the tank wall, with an uncertainty of ± 1 mm.

The breaking distance x_b was measured by manually marking the horizontal position of breaking in Tracker using the inbuilt measurement function. The dominant uncertainty is not the pixel calibration ($\sigma_l = \sigma_{\text{px}} \times 0.579$ mm/px $= 1.16$ mm) but rather the subjectivity of the visual breaking criterion. We estimate this contributes an uncertainty of ± 10 mm to each x_b measurement, as quoted in **TABLE II**.

SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

Waves in the ocean come in many shapes and sizes, but one particularly special type is the *solitary wave* — a single, smooth hump of water that travels along the surface without changing its shape. These waves were first studied in 1834 by the Scottish engineer John Scott Russell, who observed one travelling along a canal for several miles after a boat came to a sudden stop. Today, solitary waves are used as models for tsunamis, making the study of how they behave near shorelines an important practical problem.

In this experiment, we produced solitary waves in a 3 m water tank by pushing a paddle by hand. By filming the waves with a phone camera and tracking the motion of small polystyrene beads floating on the surface, we measured the speed and shape of each wave. We found that the waves produced in our tank closely matched the mathematical predictions of a theory developed by Korteweg and de Vries in 1895, confirming that our simple paddle method can reliably produce solitary waves — but only when the water is deep enough relative to the wave height.

We then placed a ramp at the end of the tank to simulate a sloping shoreline. As a solitary wave travels up a slope, it grows taller and narrower — a process called *shoaling* — until it eventually breaks. We filmed the breaking waves at seven different water depths and measured how far along the ramp each wave broke. We found that deeper water produced waves that broke further up the ramp, in agreement with the theoretical predictions of Grilli et al. (1997). However, the waves in our experiment consistently broke earlier than the theory predicted, which we believe is due to energy being lost as the wave travelled from the smooth tank floor onto the rougher surface of the ramp.